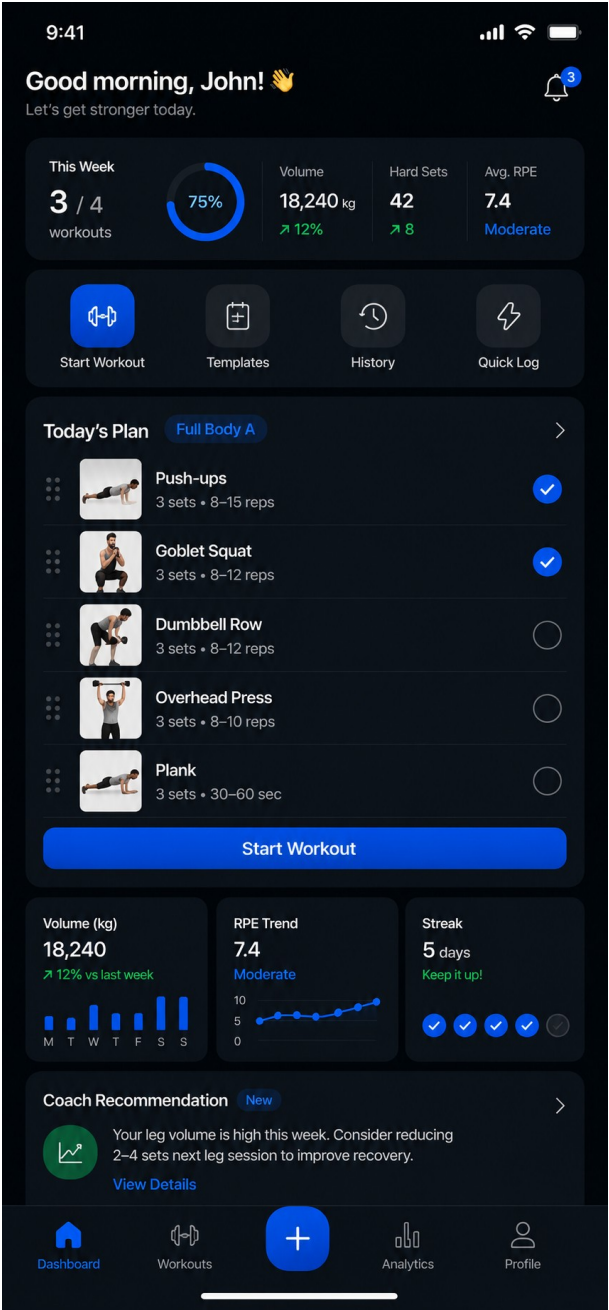


FitCoach Mobile App

Screen-to-Screen Product Requirements, UI Inventory, Swimlanes, and Test Plan

Dashboard • Active Workout • Analytics • Supporting Destinations



Version 2.0 | June 2026

1. Purpose and Scope

This document defines where every visible button, card, tab, icon, and interactive row leads in the FitCoach mobile application. It is intended to guide Flutter navigation, route naming, interaction design, QA testing, and future prototype implementation.

Interaction rule: buttons either navigate to another screen, open a modal or bottom sheet, change a local state, or run a confirmed action. No interactive element should appear tappable without a defined result.

1.1 Screen Inventory

ID	Screen	Purpose
D01	Dashboard	Daily summary, today's plan, quick actions, coach recommendation
W01	Workout Templates	Browse and start saved workout templates
W02	Workout History	Review completed and incomplete sessions
W03	Quick Log	Rapidly record an exercise, cardio session, body stat, or note
W04	Active Workout	Log sets, control rest timer, skip/edit exercises, finish session
W05	Exercise Library	Search, filter, favorite, and select exercises
W06	Exercise Details	Instructions, muscle groups, equipment, history, and logging
A01	Analytics Overview	Weekly metrics, volume, frequency, muscle balance, and RPE
A02	Progress	Exercise-specific strength, volume, RPE, and personal records
A03	Muscle Analytics	Volume and recovery by muscle group
C01	Coaching	Rule-based recommendations with optional AI explanation
N01	Notifications	Reminders, deload alerts, RPE alerts, and notification history
P01	Profile	Goals, training level, body stats, backup, export, and settings
S01	Settings	Units, appearance, security, reminders, backup, import, and app info
X01	Create Menu	Central action sheet for starting or recording activity

2. Global Navigation

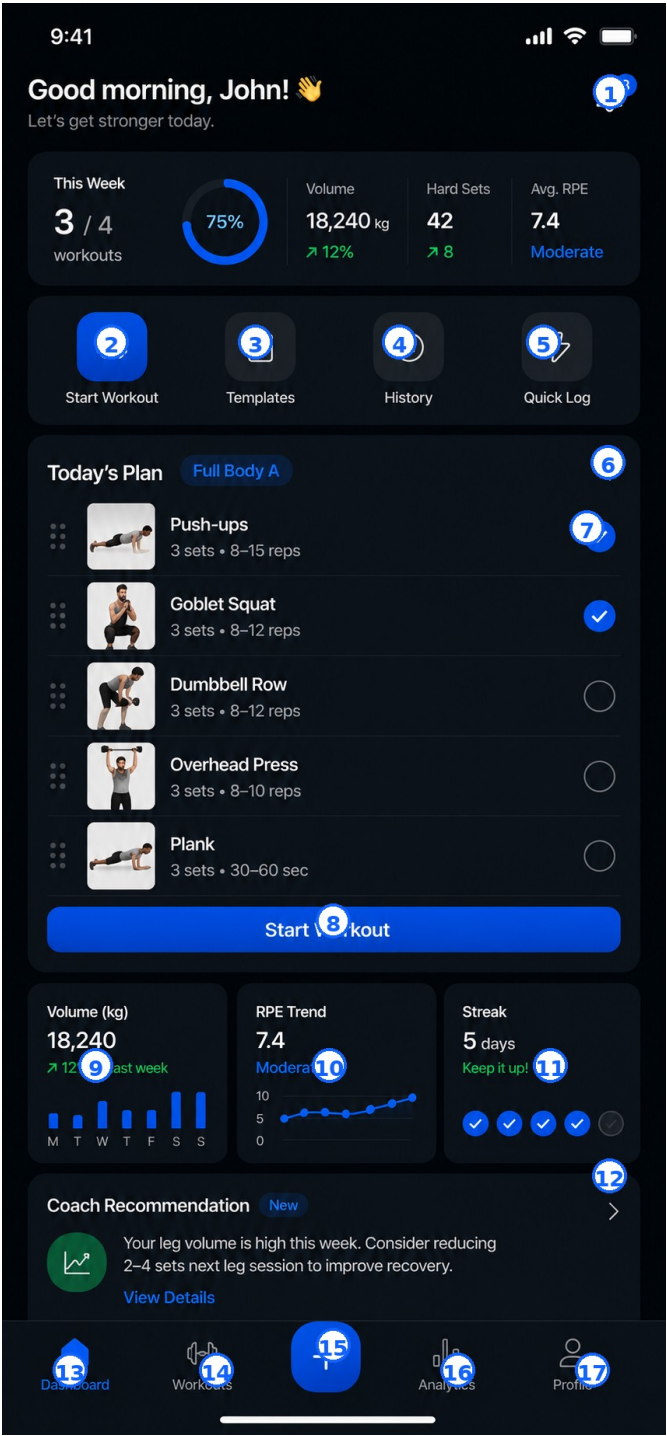
The persistent bottom navigation appears on primary screens. The active destination uses a blue icon and label. The center plus button opens a creation menu instead of replacing the current tab.

Control	Destination / Action	Behavior
Dashboard	D01 Dashboard	Returns to the latest dashboard state.
Workouts	W01 Workout Templates	Opens templates by default; preserves last selected Workouts tab when practical.
Center +	X01 Create Menu	Opens a bottom sheet with Start Workout, Quick Log, Create Template, Log Cardio, and Add Body Stat.
Analytics	A01 Analytics Overview	Opens the last selected analytics tab and date range.
Profile	P01 Profile	Opens active profile overview and account-local tools.

2.1 Recommended Flutter Routes

Screen	Route
Dashboard	/dashboard
Workout Templates	/workouts/templates
Workout History	/workouts/history
Quick Log	/quick-log
Active Workout	/workouts/session/:sessionId
Exercise Library	/exercises
Exercise Details	/exercises/:exerciseId
Analytics Overview	/analytics
Progress	/analytics/progress
Muscle Analytics	/analytics/muscles
Coaching	/coaching
Notifications	/notifications
Profile	/profile
Settings	/settings

3. Dashboard Screen (D01)



Numbered interaction map - Dashboard

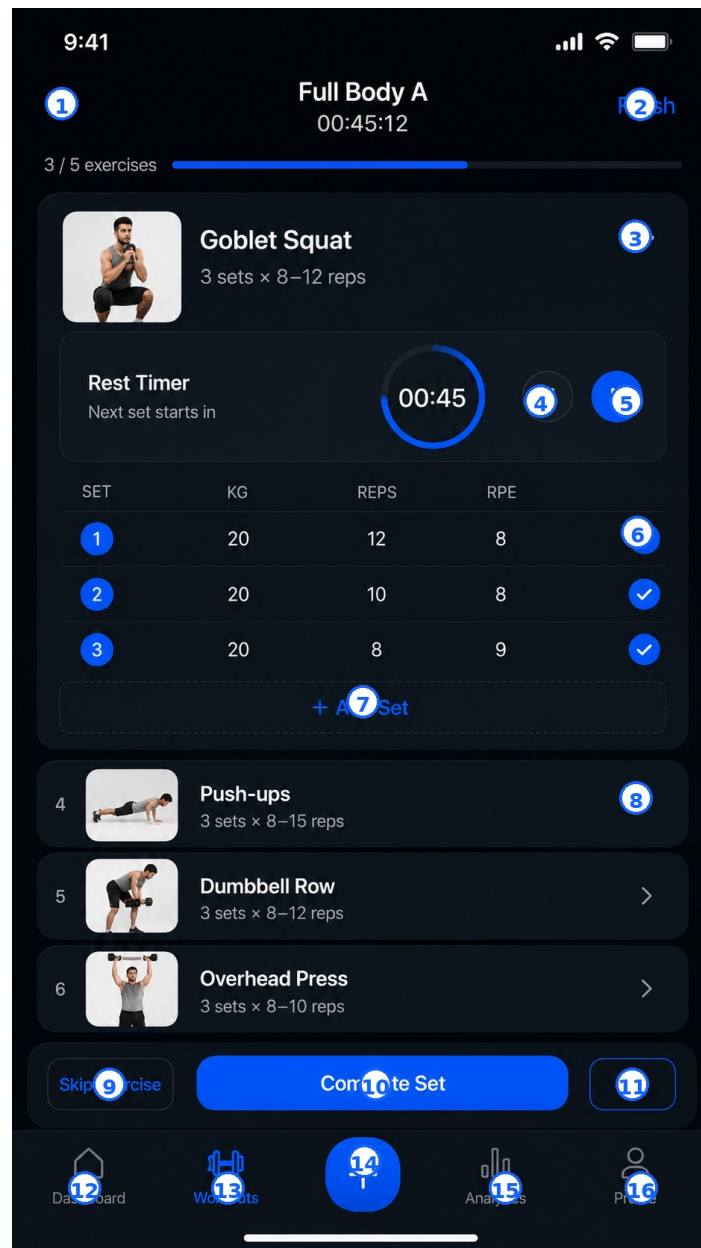
#	Button / Element	Destination	Required Behavior
1	Notification bell	N01 Notifications	Open notification center; badge shows unread count.
2	Start Workout quick action	W01 Workout Templates	Open templates with recommended template highlighted.
3	Templates quick action	W01 Workout Templates	Browse saved and default templates.
4	History quick action	W02 Workout History	Show recent sessions with

			filters.
5	Quick Log	W03 Quick Log	Open rapid logging bottom sheet or screen.
6	Today's Plan chevron	Workout Plan Details	Show planned exercises, substitutions, warm-up, and estimated duration.
7	Exercise completion circle	Local state / exercise details	Tap circle to mark planned item ready/completed; tap row to open W06.
8	Start Workout	W04 Active Workout	Create a session from today's plan and start the first incomplete exercise.
9	Volume card	A01 Analytics Overview	Open analytics with Volume metric focused.
10	RPE Trend card	A02 Progress	Open Progress with RPE Trend selected.
11	Streak card	Workout History / Streak detail	Show adherence calendar and streak rules.
12	Coach recommendation	C01 Coaching	Open recommendation details, evidence, and feedback controls.
13	Dashboard tab	D01 Dashboard	Remain on dashboard; optional scroll-to-top on repeat tap.
14	Workouts tab	W01 Workout Templates	Open workout area.
15	Center +	X01 Create Menu	Open creation bottom sheet.
16	Analytics tab	A01 Analytics Overview	Open analytics.
17	Profile tab	P01 Profile	Open profile.

3.1 Dashboard States

State	UI Change
No workout planned	Replace Today's Plan list with "Choose a workout" and a Select Template button.
Workout in progress	Primary button changes to Resume Workout and opens the existing session.
Workout completed	Show completion summary, recovery prompt, and View Session button.
No analytics data	Display onboarding cards rather than zero-value charts.
Coach unavailable	Show deterministic recommendation text; hide conversational AI action.

4. Active Workout Screen (W04)



Numbered interaction map - Active Workout

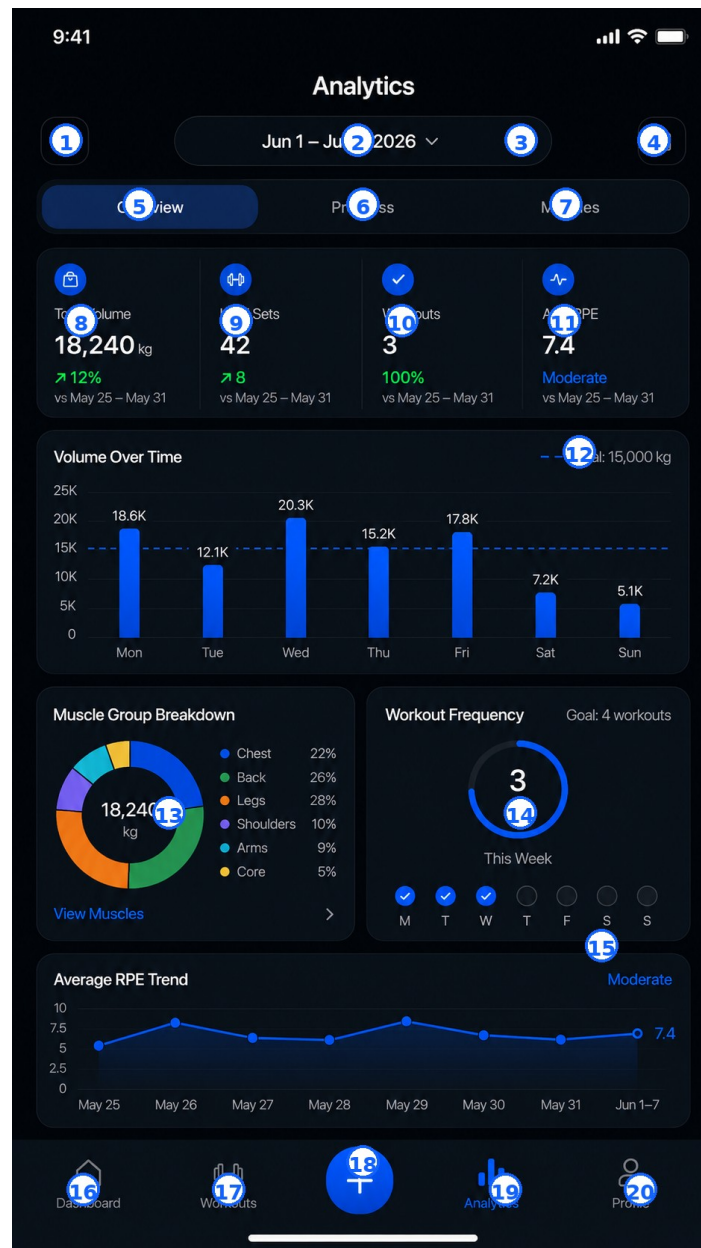
#	Button / Element	Destination / Action	Required Behavior
1	Back	Previous screen / confirmation	If unsaved changes exist, ask: Save as draft, Discard, or Continue Workout.
2	Finish	Workout Summary	Validate incomplete sets, then show finish confirmation and summary.
3	More menu	Workout actions sheet	Options: Reorder exercises, Add exercise, Replace exercise, Notes, Settings, Cancel workout.
4	Pause timer	Local timer state	Pause countdown and change control to Resume.
5	Resume/start timer	Local timer state	Start or continue countdown. Use haptic cue at zero.

6	Set completion check	Local set state	Mark set complete; save values; automatically start rest timer.
7	Add Set	Inline row	Append a set using previous set values as defaults.
8	Upcoming exercise card	Selected exercise panel	Expand the selected exercise and collapse the current one after confirmation.
9	Skip Exercise	Skip reason sheet	Require optional reason; preserve it in session history.
10	Complete Set	Local set action	Validate weight/ reps/RPE, mark current set complete, start timer, advance focus.
11	Edit	Exercise editor	Edit target sets, rep range, rest duration, notes, or substitute exercise.
12	Dashboard tab	D01 Dashboard	Warn that workout remains active; show Resume banner on dashboard.
13	Workouts tab	W01 Workout Templates	Open workout area while preserving active session.
14	Center +	X01 Create Menu	During active session, limit menu to Add Exercise, Log Note, or Log Cardio.
15	Analytics tab	A01 Analytics Overview	Open read-only analytics; keep workout timer active.
16	Profile tab	P01 Profile	Open profile; active session remains persistent.

4.1 Set Entry Rules

Field	Validation	Default
Weight	Must be ≥ 0 ; supports kg/lb conversion and bodyweight flag.	Previous completed set or last session.
Reps	Whole number ≥ 0 ; allow time/distance mode for relevant exercises.	Previous set value.
RPE	1-10, optional when user disables RPE tracking.	Blank or previous set.
RIR	0-10 when enabled; do not require both RPE and RIR.	Derived only when configured.
Completion	A set is complete only after validation and persistent save.	Unchecked.

5. Analytics Screen (A01)



Numbered interaction map - Analytics Overview

#	Button / Element	Destination / Action	Required Behavior
1	Back	Previous screen	Return without losing selected date range or tab.
2	Date-range selector	Date Range Sheet	Choose week, month, quarter, year, or custom dates.
3	Next period arrow	Next date range	Disabled when the selected range would exceed today.
4	Calendar icon	Calendar picker	Open full date picker and jump to a period.
5	Overview tab	A01 Analytics Overview	Display summary metrics and charts.
6	Progress tab	A02 Progress	Display exercise-level trends and records.
7	Muscles tab	A03 Muscle Analytics	Display muscle-group volume and recovery.

8	Total Volume metric	Volume detail	Show session and exercise contributors to total load.
9	Hard Sets metric	Hard-set detail	Show hard sets by muscle and workout.
10	Workouts metric	Workout History filtered	Open completed workouts in selected range.
11	Average RPE metric	RPE detail	Show distribution, exercise contributors, and fatigue warnings.
12	Volume chart	Chart detail / tooltip	Tap bars for exact values; long press for comparison.
13	Muscle Group Breakdown	A03 Muscle Analytics	Open muscle details for the selected segment.
14	Workout Frequency	Adherence detail	Show planned versus completed sessions and streaks.
15	Average RPE Trend	A02 Progress - RPE	Open larger RPE chart and session list.
16	Dashboard tab	D01 Dashboard	Open dashboard.
17	Workouts tab	W01 Workout Templates	Open workout area.
18	Center +	X01 Create Menu	Open creation bottom sheet.
19	Analytics tab	A01 Analytics Overview	Repeat tap scrolls to top.
20	Profile tab	P01 Profile	Open profile.

5.1 Analytics Empty and Error States

Condition	Response
No workouts in range	Show “No activity yet” with Start Workout and Change Date Range actions.
Only one session	Show totals; label trend charts as insufficient data rather than drawing misleading lines.
Missing RPE	Calculate other metrics and label RPE card “Not recorded”.
Data calculation error	Show retry action and diagnostics reference; do not display stale totals without a label.
Deload week	Overlay a Deload badge and compare against the previous training week separately.

6. Supporting Destination Screens

6.1 Workout Templates (W01)

Button / Element	Destination / Action	Behavior
Template card	Template Details	Preview exercises, sets, duration, equipment, and recent performance.
Start	Active Workout	Create a session and open the first exercise.
Create Custom Workout	Template Builder	Add exercises, targets, ordering, and rest rules.
History tab	Workout History	Switch within Workouts area.
Favorites tab	Favorite Templates	Filter templates marked as favorites.
Search/filter	Filtered list	Search by name, muscle group, equipment, or duration.

6.2 Workout History (W02)

Button / Element	Destination / Action	Behavior
Session row	Session Details	Show all exercises, sets, notes, duration, volume, and RPE.
Repeat Workout	Active Workout	Clone the session as a new workout.
Edit Session	Session Editor	Allow corrections while maintaining an audit timestamp.
Delete	Confirmation dialog	Require confirmation and offer undo where possible.
Filter	Filter sheet	Filter by date, template, exercise, completion state, and tag.

6.3 Quick Log (W03)

Button / Element	Destination / Action	Behavior
Strength Exercise	Exercise picker	Choose exercise and log one or more sets.
Cardio	Cardio form	Activity, duration, distance, heart rate, and effort.
Body Stat	Body stat form	Weight, measurements, recovery, or custom metric.
Note	Daily note form	Record soreness, sleep, stress, energy, or free text.
Save	Previous screen	Validate and persist; show success feedback.

6.4 Exercise Library and Details (W05/W06)

Button / Element	Destination / Action	Behavior
Search	Filtered results	Live search by exercise name and alias.
Category chips	Filtered results	All, Push, Pull, Legs, Core, Cardio, Mobility.
Favorite star	Local favorite state	Toggle favorite and sync to template picker.
Exercise row	Exercise Details	Open instructions, muscles, equipment, and history.
Log Exercise	Quick Log / Active Workout	Context-aware: add to active workout or create standalone entry.
History tab	Exercise history	Show charts, personal records, and recent sets.

6.5 Coaching (C01)

Button / Element	Destination / Action	Behavior
Recommendation card	Recommendation Details	Show rule key, evidence, confidence, and proposed action.
Accept	Apply action	Update the plan only after explicit confirmation.
Dismiss	Dismiss reason sheet	Collect reason without disabling safety rules.
Thumbs up/down	Feedback record	Rate explanation usefulness; do not directly override safety logic.
Tell me more	AI explanation	Request optional Ollama explanation of structured evidence.

6.6 Profile and Settings (P01/S01)

Button / Element	Destination / Action	Behavior
Goal	Goal editor	Choose strength, muscle, endurance, general fitness, or custom goal.
Training level	Level editor	Beginner, intermediate, advanced, with explanation.
Body Stats	Body stats history	View and add measurements.
Training Blocks	Block calendar	Review training/deload weeks and upcoming schedule.
Personal Records	Records list	Filter by exercise and record type.
Backup & Restore	Backup manager	Create, verify, restore, and export local backup.
Export Data	Export wizard	CSV/JSON/PDF summary options.
Settings gear	Settings	Open units, notifications, privacy, appearance, and data controls.

7. End-to-End Screen Flows

User Goal	Screen-to-Screen Path
Start today's workout	D01 Dashboard → Start Workout → W04 Active Workout → Finish → Workout Summary → D01 Dashboard
Choose a different template	D01 Dashboard → Templates → W01 → Template Details → Start → W04
Quickly log one exercise	Any primary screen → Center + → X01 → Quick Log → W03 → Exercise Picker → Save → Previous screen
Review progress	D01 or bottom nav → A01 Analytics → Progress tab → A02 → Exercise selector → Exercise trend detail
Inspect muscle balance	A01 Analytics → Muscles tab or donut segment → A03 Muscle Analytics → Muscle detail
Respond to coaching	D01 Coach Recommendation → C01 → Recommendation Details → Accept/Dismiss → Updated plan
Recover interrupted workout	App launch → D01 Resume banner → W04 restored session → Continue or Finish
Change profile preferences	Bottom Profile → P01 → Settings → S01 → update preference → Save → P01

7.1 Navigation Guard Rules

Situation	Required Guard
Leaving an active workout	Keep session persistent. Confirm only when user explicitly cancels or discards.
Deleting data	Use confirmation dialog with the exact object name and consequence.
Applying coach changes	Show the change before it modifies targets or schedule.
Switching profile	Close current database connection; block switch during unsaved workout or offer save-as-draft.
Opening unavailable AI	Fall back to deterministic explanation and show local-model status.
Repeated rapid taps	Debounce actions that create sessions, logs, backups, or exports.

8. Button QA and Acceptance Criteria

Requirement	Acceptance Criterion
Tap target	Every primary control has a minimum 44×44 logical pixel hit area.
Feedback	Every tap gives immediate visual, haptic, or progress feedback.
Loading	Actions longer than 300 ms show progress; destructive buttons are disabled during submission.
Persistence	Workout and set changes survive navigation and app restart.
Back behavior	System back and UI back produce the same guarded result.
Accessibility	Icons have semantic labels; color is not the only status indicator.
Offline operation	All core routes and actions work without internet.
Error recovery	Errors preserve user-entered values and provide a retry path.
Deep links	Routes with IDs handle missing or deleted records with a safe not-found state.
Analytics integrity	Every displayed metric can be traced to underlying sessions and date filters.

8.1 Test Case Template

Field	Example
Test ID	NAV-D01-008
Starting screen	Dashboard
Control	Start Workout
Precondition	Today's plan exists and no workout is active
Expected route	/workouts/session/{newSessionId}
Expected result	Session is created once; first incomplete exercise is focused; timer is stopped.
Failure handling	Display recoverable error; do not create duplicate sessions.

9. Actors and Responsibilities

The system is local-first. The primary actor is the athlete using the mobile app. Supporting actors are internal app components or optional local services that participate in the workflow.

ID	Actor	Responsibilities
ACT-01	Athlete / Primary User	Creates profiles; selects plans; logs workouts and cardio; reviews analytics; responds to coaching; manages backups and preferences.
ACT-02	Secondary Local Profile	Uses a separate local profile and database on the same device. Cannot view or modify another profile without switching.
ACT-03	Flutter Mobile App	Renders screens, validates input, manages navigation and local UI state, schedules notifications, and communicates with the API.
ACT-04	FastAPI Backend	Applies business rules, persists transactions, computes analytics, manages profiles, and returns structured coaching recommendations.
ACT-05	DuckDB Profile Store	Stores exercises, templates, sessions, sets, feedback, alerts, training blocks, and derived analytics for the active profile.
ACT-06	Notification Scheduler	Schedules local reminders, streak notices, deload alerts, RPE alerts, and post-session prompts.
ACT-07	Rule-Based Coach	Generates deterministic suggestions using volume, RPE, progression, cardio load, readiness, and deload rules.
ACT-08	Optional Ollama Service	Converts structured recommendations into conversational explanations. It cannot alter rule outputs or prescribe medical treatment.
ACT-09	File System / Backup Service	Creates, verifies, restores, imports, and exports profile files and user-selected data formats.
ACT-10	Developer / QA Tester	Implements routes and requirements, executes test cases, validates accessibility, and verifies calculations and data integrity.

10. Functional Requirements

Priority: Must = required for MVP; Should = important after MVP stabilization; Could = optional enhancement.

ID	Priority	Area	Requirement
FR-001	Must	Profiles	Create, rename, archive, switch, export, restore, and permanently delete a profile after confirmation.
FR-002	Must	Profiles	Store each profile in an isolated local database and persist the active profile in app configuration.
FR-003	Must	Exercises	Search and filter the exercise library by name, movement, muscle group, equipment, and favorite status.
FR-004	Must	Templates	Create, edit, duplicate, favorite, reorder, archive, and start workout templates.
FR-005	Must	Workout Logging	Start, pause, resume, save as draft, cancel, and finish a workout session.
FR-006	Must	Workout Logging	Record weight, reps, duration, distance, RPE or RIR, warm-up status, notes, and completion state per set.
FR-007	Must	Workout Logging	Copy prior-session values, add or remove sets, reorder or substitute exercises, and skip with an optional reason.
FR-008	Must	Timer	Start the configured rest timer after set completion and provide pause, resume, add time, skip, haptic, and sound controls.
FR-009	Must	Persistence	Persist every completed set immediately and recover an active or draft workout after app restart.
FR-010	Must	History	Display session history and details; allow repeat, correct, export, and confirmed deletion.
FR-011	Must	Analytics	Calculate volume, hard sets, workout frequency, average RPE, muscle distribution, and exercise progress for selected periods.
FR-012	Must	Analytics	Provide drill-down from each metric and chart to the source sessions and sets.
FR-013	Must	Coaching	Generate deterministic structured recommendations and show rule evidence, severity, and proposed action.
FR-014	Should	Coaching	Collect accept, dismiss, reason, and usefulness feedback without allowing feedback to disable safety rules.
FR-015	Should	Cardio	Log running, cycling, rowing, swimming, duration, distance, heart rate, and calculated cardio stress.
FR-016	Should	Recovery	Capture optional sleep, soreness, stress, energy, and pain flags and use them only in transparent readiness rules.
FR-017	Should	Deloads	Schedule training and deload

			blocks and suppress inappropriate progression recommendations during deloads.
FR-018	Must	Notifications	Schedule, edit, disable, and cancel local workout, streak, deload, RPE, and cool-down reminders.
FR-019	Must	Data	Export CSV/JSON summaries and create verified backups that can be restored without data loss.
FR-020	Should	Import	Import validated CSV and wearable files with preview, duplicate detection, error reporting, and rollback.
FR-021	Could	AI	Use Ollama to explain a structured recommendation; the app must function fully when the model is unavailable.
FR-022	Must	Settings	Manage units, appearance, accessibility, timer defaults, notifications, backup location, and analytics preferences.
FR-023	Must	Safety	Show exercise-safety guidance and route pain or concerning symptom inputs to a stop-and-seek-help message rather than workout advice.
FR-024	Must	Auditability	Record created, updated, and source timestamps for corrected sessions, imported data, rule feedback, and restored backups.

11. Non-Functional and UI Requirements

ID	Category	Requirement
NFR-001	Offline	All core workout, history, analytics, profile, and rule-based coaching functions shall work without internet access.
NFR-002	Performance	Primary screens should render usable content within 1 second on a supported device; set-save feedback should appear within 300 ms.
NFR-003	Reliability	A session save shall be atomic. A failed write must not produce a partially completed session.
NFR-004	Recovery	The app shall recover the last active session after process termination and identify unsynced or corrupted draft state.
NFR-005	Accessibility	Support screen readers, semantic labels, text scaling, 44 x 44 logical-pixel targets, visible focus, and status indicators beyond color.
NFR-006	Contrast	Text and essential controls shall meet WCAG AA contrast targets in dark and light themes.
NFR-007	Privacy	Personal workout and health-related data shall stay local unless the user explicitly exports it.
NFR-008	Security	Optional PIN or device biometrics shall protect profile access; secrets shall not be stored in plaintext logs.
NFR-009	Integrity	Analytics totals must be reproducible from stored sessions using the selected date range and active profile.
NFR-010	Maintainability	Flutter features, repositories, API clients, rule services, and models shall be modular and covered by automated tests.
NFR-011	Compatibility	Layouts shall adapt to common phone widths and portrait orientation without clipping, overlap, or horizontal scrolling.
NFR-012	Observability	Local rotating logs and a diagnostics page shall expose app, API, database, schema, model, import, and backup status without logging sensitive notes.
NFR-013	Localization	Dates, numbers, units, and time formats shall use locale-aware formatting.
NFR-014	AI Safety	LLM output shall be labeled as explanatory, generated from structured evidence, and ignored when it conflicts with deterministic rules.
NFR-015	Backup	Before a destructive schema migration or restore, the system shall create and verify a recovery copy.

11.1 UI Component Requirements

ID	Component	Requirement
UI-001	Global App Bar	Show screen title, contextual actions, and safe back navigation. Do not display unavailable controls.
UI-002	Bottom Navigation	Dashboard, Workouts, central Create action, Analytics, and Profile remain consistent on primary screens.

UI-003	Cards	Cards must have a clear title, primary value, secondary context, and an explicit tap result when interactive.
UI-004	Forms	Show units, allowed ranges, prior values, inline validation, and preserve values after recoverable errors.
UI-005	Charts	Provide labels, exact-value tooltips, accessible summaries, empty states, and source-data drill-down.
UI-006	Destructive Actions	Use clear destructive styling, name the affected record, explain consequences, and require confirmation.
UI-007	Loading	Use skeletons for initial loads and progress indicators for actions; prevent duplicate submissions.
UI-008	Errors	State what failed, preserve user work, provide retry or recovery, and include a diagnostics reference when appropriate.
UI-009	Notifications	Request permission contextually and explain how each notification type can be changed.
UI-010	Active Workout	Keep the primary set action reachable near the bottom; do not hide the current exercise, set fields, or timer behind navigation.

12. Actor Swimlanes

The swimlanes below define responsibility boundaries and expected handoffs. Each row is a sequential step; a dash indicates that the actor has no action at that step.

12.1 Start and Complete a Workout

Step	Athlete	Flutter App	FastAPI	DuckDB	Notifications	Rule Coach
1	Athlete taps Start Workout.	Validates navigation and sends start request.	Creates session transaction.	Inserts session and planned exercises.	-	-
2	Reviews first exercise.	Renders prior values and current targets.	Returns exercise/session payload.	Reads template and history.	-	-
3	Enters set values and taps Complete Set.	Validates fields and shows pending state.	Validates business rules and saves set.	Commits set atomically.	-	-
4	Waits or controls rest timer.	Starts local timer and schedules cue.	-	-	Provides local alert/haptic at zero.	-
5	Repeats sets/exercises.	Persists each completed set and progress state.	Updates session totals and detects alerts.	Stores set records and alert state.	May surface timer/reminder cues.	Evaluates rules from structured data.
6	Taps Finish and confirms.	Shows completion summary request.	Validates session and finalizes totals.	Commits completed status.	Cancels obsolete workout reminder.	Generates post-session recommendations.
7	Reviews summary.	Displays volume, hard sets, RPE, records, and next action.	Returns computed summary.	Reads finalized session.	May schedule recovery/deload reminder.	Returns rule evidence and recommendation.

12.2 View Analytics and Drill Down

Step	Athlete	Flutter App	FastAPI	DuckDB	Rule Coach
1	Opens Analytics and chooses a date range.	Sends profile and range query.	Validates range and requests metrics.	Aggregates source sessions/sets.	-
2	Views summary cards and charts.	Renders values, labels, empty states, and comparison.	Returns totals and chart series.	Reads derived views or computes query.	-
3	Taps Hard Sets or a chart bar.	Sends drill-down query with metric and selection.	Returns contributors and calculation metadata.	Reads matching sessions and sets.	-
4	Selects a source session.	Navigates to session details.	Returns full session detail.	Reads exact records.	-
5	Changes or clears the range.	Updates controls and refreshes state.	Recomputes using new period.	Executes period-filtered query.	-

12.3 Coaching Recommendation and Feedback

Step	Athlete	Flutter App	FastAPI	DuckDB	Rule Coach	Ollama
1	Opens Coaching.	Requests current suggestions.	Builds structured context.	Reads recent volume, RPE, readiness, cardio, and blocks.	Evaluates deterministic rules.	-
2	Reads recommendation.	Displays action, evidence, severity, and source data.	Returns structured result.	-	Returns rule output.	Optionally writes an explanation without changing values.
3	Taps Accept.	Shows exact proposed change and confirmation.	Applies approved plan change.	Stores updated target and audit record.	Re-evaluates dependent rules.	-
4	or taps Dismiss and reason.	Sends feedback without applying action.	Stores feedback event.	Inserts feedback and context snapshot.	Safety rule remains active.	May be rated separately for explanation quality.

12.4 Backup and Restore

Step	Athlete	Flutter App	FastAPI	DuckDB	File/Backup Service
1	Chooses Create Backup.	Shows destination and scope.	Quiesces writes and requests backup.	Checkpoints active database.	Copies files to temporary package.

2	Confirms.	Shows progress and blocks duplicate action.	Creates manifest and checksum.	Provides schema/profile metadata.	Writes package and verifies checksum.
3	Selects Restore and backup file.	Shows preview, date, version, and warnings.	Validates compatibility and creates pre-restore backup.	Closes active connection.	Restores to temporary location and verifies.
4	Confirms restore.	Restarts data state and shows result.	Switches to restored database and runs migrations.	Opens restored data and validates counts.	Keeps rollback copy until success.

13. Function Inventory

ID	Module	Functions
FN-001	Profile Management	createProfile, renameProfile, switchProfile, archiveProfile, deleteProfile, listProfiles
FN-002	Template Management	createTemplate, updateTemplate, duplicateTemplate, reorderTemplateExercises, archiveTemplate, favoriteTemplate
FN-003	Exercise Library	searchExercises, filterExercises, getExerciseDetails, toggleFavoriteExercise, createCustomExercise
FN-004	Session Lifecycle	startSession, resumeSession, saveDraft, finishSession, cancelSession, recoverActiveSession
FN-005	Set Logging	addSet, updateSet, completeSet, uncompleteSet, deleteSet, copyPreviousSet
FN-006	Exercise in Session	addExercise, replaceExercise, reorderExercises, skipExercise, addExerciseNote
FN-007	Rest Timer	startTimer, pauseTimer, resumeTimer, addTimerSeconds, skipTimer, issueTimerCue
FN-008	Workout History	listSessions, getSession, repeatSession, correctSession, deleteSession, exportSession
FN-009	Analytics	getWeeklySummary, getVolumeSeries, getHardSets, getFrequency, getRpeTrend, getMuscleBreakdown, drillDownMetric
FN-010	Progress	getExerciseProgress, calculateEstimated1RM, listPersonalRecords, getBestSets, comparePeriods
FN-011	Cardio	logCardio, importCardio, calculateCss, getRollingCardioLoad
FN-012	Readiness	logReadiness, calculateReadiness, flagPain, getRecoveryTrend
FN-013	Coaching	getSuggestions, getSuggestionEvidence, acceptSuggestion, dismissSuggestion, rateExplanation
FN-014	Deloads	createTrainingBlock, scheduleDeload, triggerReactiveDeload, suppressProgressionDuringDeload
FN-015	Alerts	detectRpeTrend, createAlert, clearAlert, listAlerts
FN-016	Notifications	requestPermission, scheduleReminder, updateReminder, cancelReminder, listNotificationHistory
FN-017	Import/Export	previewImport, validateImport, deduplicateImport, commitImport, exportCsv, exportJson
FN-018	Backup/Restore	createBackup, verifyBackup, listBackups, restoreBackup, rollbackRestore
FN-019	Settings	getSettings, updateSettings, resetLearnedPreferences, setUnits, setTheme
FN-020	Diagnostics	getHealthStatus, getSchemaVersion, getModelStatus, getLastBackup, exportDiagnosticLog

14. Test Plan and Test Cases

Test levels include unit, widget, API, integration, persistence/recovery, accessibility, and exploratory testing. Critical flows must be tested on at least one small and one large phone viewport.

14.1 Entry and Exit Criteria

Type	Criteria
Entry	Requirements and routes are approved; test data is seeded; migrations are current; build installs; major dependencies are available or have documented fallback states.
Exit	All Must requirements pass; no unresolved critical/high defects; critical flows pass on supported platforms; analytics reconciliation passes; backup/restore is verified; accessibility blockers are resolved.

ID	Priority	Area	Test	Precondition	Action	Expected Result
TC-001	Critical	Dashboard	Start workout from today plan	Planned workout exists; no active session	Tap Start Workout	Exactly one new session opens on first exercise; draft is persisted.
TC-002	Critical	Workout	Recover interrupted session	Active session contains one completed set	Terminate and relaunch app	Dashboard shows Resume; session restores values, order, elapsed state, and completion.
TC-003	High	Workout	Reject invalid set input	Active set visible	Enter negative weight or RPE 11; tap Complete Set	Inline error appears; set is not saved; valid fields remain.
TC-004	Critical	Workout	Complete set atomically	Valid set values entered	Tap Complete Set once and rapidly tap again	One set completion is stored; timer starts once; no duplicate write.
TC-005	High	Workout	Skip exercise with reason	Active exercise not completed	Tap Skip Exercise; choose Equipment unavailable	Exercise is skipped; reason appears in summary and history.
TC-006	High	Workout	Finish with incomplete items	Some exercises incomplete	Tap Finish	Confirmation lists incomplete items and offers Continue or Finish Anyway.
TC-007	Critical	History	Correct completed session	Completed session exists	Edit reps and save	Totals recompute; updated timestamp and audit source are stored.
TC-008	High	Analytics	Reconcile total volume	Known seeded sessions in selected week	Open weekly analytics	Displayed volume equals sum(weight x reps) under configured calculation rules.
TC-009	High	Analytics	Missing RPE handling	Sessions have no RPE	Open analytics	Other metrics render; RPE shows Not recorded; no fake zero trend.
TC-010	High	Analytics	Drill down hard sets	Hard sets exist	Tap Hard Sets card	Contributor list totals equal card value and links open source sessions.
TC-011	High	Analytics	Empty period	No sessions in date range	Select empty range	No activity state appears with Start Workout and Change Range actions.
TC-012	High	Coaching	Rule output remains deterministic	Seed known volume/RPE context	Request suggestion twice	Rule key, action, evidence, and thresholds match; prose may vary only in optional AI explanation.
TC-013	Critical	Safety	Pain flag response	Readiness form available	Mark chest pain or dizziness	App stops workout recommendation

						and shows urgent safety guidance; LLM is not consulted.
TC-014	High	Coaching	Dismiss does not disable safety rule	High fatigue rule active	Dismiss recommendation	Feedback is stored; active safety/fatigue rule can still fire.
TC-015	High	Profiles	Switch profile isolation	Two profiles have different histories	Switch profile	Only selected profile data appears; connections close cleanly; no cross-profile records.
TC-016	Critical	Backup	Verified backup and restore	Profile has sessions/templates/settings	Create backup, add data, restore backup	Restored counts and checksums match backup; pre-restore rollback copy exists.
TC-017	High	Import	Duplicate detection	CSV contains existing and new sessions	Preview and import	Existing records are flagged; user chooses skip/merge; transaction can roll back on error.
TC-018	Medium	Notifications	Reminder edit and cancel	Permission granted	Schedule, change time, then disable reminder	OS schedule reflects changes and no disabled reminder fires.
TC-019	High	Accessibility	Screen reader navigation	Screen reader enabled	Navigate dashboard and workout	Controls have meaningful labels, reading order is logical, charts have summaries, and state is not color-only.
TC-020	High	Responsive UI	Small phone viewport	Viewport near supported minimum	Open all primary screens and text scale to 200%	No clipped primary actions, overlapping text, or inaccessible horizontal content.
TC-021	High	Offline	Core workflow offline	Network disabled	Start/log/finish workout and open analytics	All core actions succeed; optional Ollama shows unavailable fallback.
TC-022	Medium	Timer	Timer background behavior	Rest timer running	Lock device or navigate away and return	Remaining time is correct from timestamps; cue occurs once.
TC-023	High	Migration	Upgrade old profile schema	Backup of prior schema version	Launch new app version	Backup is created; migrations apply once; data counts and key relationships remain valid.
TC-024	Medium	Settings	Unit conversion	Existing kg records	Change display units to lb	Stored canonical values remain unchanged; display and entry convert consistently.
TC-025	High	Navigation	Back guard during workout	Unsaved active workout	Use UI and system back	Both show Save Draft, Discard, Continue; selecting each performs the documented result.

14.2 Automated Test Coverage Targets

Layer	Minimum Target	Required Focus
Rule engine	90% line/branch	Threshold boundaries, conflicting rules, deload suppression, safety invariants.
Backend services	80% line	Transactions, migrations, profile isolation, analytics reconciliation, import and backup.
Flutter units/widgets	80% critical modules	Validation, state restoration, navigation guards, empty/error/loading states.
Integration	All critical flows	Start-to-finish workout, recovery,

		analytics drill-down, profile switch, backup/restore.
Accessibility	All primary screens	Semantics, focus order, text scale, contrast, non-color status, touch target size.

15. Requirements Traceability Matrix

Requirements	Screens	Functions	Tests
FR-001, FR-002	P01/S01	FN-001	TC-015, TC-016
FR-003	W05/W06	FN-003	Search/filter widget and API tests
FR-004	W01	FN-002	Template create/edit/start tests
FR-005-FR-009	W04	FN-004-FN-007	TC-001-TC-006, TC-022, TC-025
FR-010	W02	FN-008	TC-007
FR-011, FR-012	A01-A03	FN-009, FN-010	TC-008-TC-011
FR-013, FR-014, FR-021	C01	FN-013	TC-012, TC-014, TC-021
FR-015-FR-017	W03/C01/Calendar	FN-011, FN-012, FN-014	Cardio, readiness, and deload rule tests
FR-018	N01/S01	FN-016	TC-018
FR-019, FR-020	P01/S01	FN-017, FN-018	TC-016, TC-017
FR-022	S01	FN-019	TC-024 plus theme/accessibility tests
FR-023	Readiness/Workout/Coaching	FN-012, FN-013	TC-013
FR-024	History/Import/Backup	FN-008, FN-017, FN-018	Audit timestamp and source assertions
NFR-001-NFR-015	All primary screens	Cross-cutting	TC-002, TC-016, TC-019-TC-025

16. Definition of Done

Area	Done When
Requirements	All Must functional and non-functional requirements have acceptance evidence.
Navigation	Every visible interactive element has a defined route, modal, local state, or confirmed action.
Data	Session writes are atomic; profile isolation and analytics reconciliation pass.
Recovery	Active workout, migration, import, backup, and restore recovery paths are verified.
Safety	Pain and concerning symptom workflows bypass AI and display appropriate stop guidance.
Accessibility	Primary screens pass semantics, focus, contrast, scaling, and touch-target checks.
Testing	Critical automated and manual tests pass with no unresolved critical/high defects.
Documentation	Routes, API contracts, schema migrations, rules, calculations, and release notes are current.